What Would It Take for Physics to Favor a Theory of Time?

Bridge Premises, Records, and Conditional Constraints

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Abstract

Physical theories constrain temporal metaphysics, but the target of a constraint must be specified. Presentism, eternalism, and accounts of becoming do not, by their labels alone, determine experimental probabilities. This draft argues for a contrast-sensitive burden of proof: an inference from physics to a temporal ontology should identify an admissible physical realization, the bridge premises connecting its structures to existence or passage, and the alternative that the evidence or argument discriminates against. We distinguish physical incompatibility, empirical discrimination, and representation equivalence. Special-relativistic simultaneity and relational quantum clocks illustrate why these inferential forms cannot substitute for one another. An elementary finite-history example makes record preservation precise while exposing its limits. The position is neither that physics is irrelevant to metaphysics nor that empirical equivalence makes ontologies identical. The proposed framework and examples are expository and argumentative; no mathematical or philosophical priority, new physical prediction, or preferred temporal ontology is established here.

Draft status. Citations vary in reading depth; the companion `SOURCE_AUDIT.md` records those limits. The manuscript does not report a completed systematic review, a proof-assistant verification, or an original result that has passed the project's independent novelty audit.

1 The question is contrastive

What would it mean for physics to favor eternalism over presentism, or local becoming over a moving spotlight? At least three answers are possible. A view might contradict a well-supported physical structure. A realized model might predict observations more successfully than a rival. Or the physical theory might provide an explanation whose ontological commitments are better understood in one way than another. These are serious routes to philosophical justification, but they have different premises and different conclusions.

The danger is not merely an insufficient number of citations. It is an inference in which a result answering one question is taken to answer another: the absence of a distinguished simultaneity relation becomes the nonexistence of passage; the representation of an entire history becomes determinism; or an equivalence of relational dynamics becomes the elimination of fundamental time. A dense catalogue of physical theories cannot repair that inferential gap without identifying the premises that cross it.

Our methodological starting point is physics-constrained but non-reductive. Empirical adequacy matters greatly, while meaningful claims about grounding, existence, and explanation are not defined away because they lack a direct meter reading. This starting point is itself a philosophical commitment, not a theorem of physics. The breadth of the field is usefully represented by Callender

(2011); the contrasting programs of Callender (2017) and Maudlin (2007) warn against treating one account of passage as the neutral interpretation of scientific practice.

Our thesis is therefore a burden-of-proof claim:

> An argument that physics favors a temporal ontology must identify the realization and bridge premises under which the physical result bears on that ontology, and specify the contrast that the result actually supports.

This does not require every philosophical argument to predict a new measurement. It requires an argument to say whether its force is empirical, logical, explanatory, or a combination, and to preserve that scope at its conclusion.

2 What is being compared?

2.1 Independent questions

For this draft, presentism restricts unrestricted concrete existence to what is present; eternalism admits past and future entities as well as present ones. Growing-block views retain the past and present while rejecting future entities; moving-spotlight accounts combine a complete temporal ontology with objectively distinguished presentness. These are schematic families, not complete theories. Direct defenses differ substantially: compare Bourne (2006), Correia and Rosenkranz (2018), and Cameron (2015). Their detailed replies have not yet been fully audited for this manuscript.

Several further questions must remain separate. Are tensed facts irreducible? Does something objectively become present, existent, or definite? Do laws and complete appropriate data select one history? How do objects persist? Is time fundamental? Which asymmetries need explanation? An eternalist can accept indeterministic laws. A four-dimensional spacetime does not by itself make ordinary objects temporal worms. Knowing that an event is now is not automatically knowing an extra objective spotlight property.

These distinctions are not concessions invented to protect a preferred position. McTaggart's argument already differentiates temporal order, change, and A-determinations (McTaggart, 1908); its contested premises cannot be replaced by the phrase "time is an illusion." Likewise, Skow (2015) distinguishes robust from less demanding senses of passage. No result about clock rates alone establishes which sense is physically real.

2.2 Realizations, not bare labels

Represent a comparison target schematically as

$$M = (P, O, B, A, S; \mathcal{E}),$$

where P specifies a physical model and dynamics, O an ontological interpretation, B the bridge premises connecting physical structure to ontological claims, A auxiliary and boundary assumptions, S semantic conventions, and $\mathcal{E}$ the allowed preparations, observations, and interventions. The tuple is bookkeeping rather than a new physical formalism; its components need not be independent.

An ontology may have several proposed realizations. Write $\mathcal{R}(O)$ for the specified admissible class, where admissibility must be defended independently of rescuing a view after each objection. An empirical likelihood is supplied by a realized package and an experimental protocol, not simply by its label. This observation does not make ontological hypotheses impossible to assess: one may compare independently justified realization classes or marginalize over auxiliaries with explicit priors. It blocks only the unannounced move from a result for one realization to a verdict on every realization.

Constraining admissibility without settling the dispute by fiat. Before inspecting a proposed discriminator, register at least four conditions: (i) mathematical consistency and empirical adequacy in the declared regime; (ii) fidelity to the ontology's explicitly stated constitutive commitments, rather than merely retaining its name; (iii) a specified dynamics, probability model and operational interface, including proposed auxiliaries; and (iv) an independent motivation for additional structure, assessed against the same standards used for rivals. The first conditions constrain which models enter the comparison. Simplicity, explanatory contribution and costs of primitive structure then help rank the survivors; they are not covert empirical vetoes.

The proponent bears the burden of describing a realization and defending its bridges; the exclusion advocate bears the stronger burden of defending the scope of the excluded class. Neither party may change those standards silently after a counterexample. Where reasonable standards disagree, report conditional verdicts under both rather than call one class uniquely admissible. This does not mechanically settle abduction, but it makes auxiliary rescue and asymmetric demands on rivals inspectable.

3 Three different inferential burdens

3.1 Exclusion

A physical incompatibility argument has a form such as

$$O \wedge B \wedge A \implies F, \qquad P \implies \neg F.$$

It follows that P excludes the conjunction $O \wedge B \wedge A$. Concluding $\neg O$ requires independent support for the indispensable bridge and auxiliary premises, or an argument that every admissible realization of O encounters the conflict. Otherwise the result identifies a conditional cost: retain the ontology only by revising the disputed structure or bridge.

This is not the claim that a bare ontology can always be rescued. A bridge may be constitutive of a precisely defined view, or its removal may destroy the explanation that made the view attractive. That stronger argument should be made rather than hidden in a generic ontology label.

3.2 Empirical discrimination

Empirical discrimination requires different probability assignments to a common experimental question. At fixed realized packages, if

$$p(D \mid M_1, e) = p(D \mid M_2, e)$$

for every outcome event D and protocol $e \in \mathcal{E}$, those observations cannot discriminate the packages by their likelihoods. This statement is restricted to the declared protocol class, not all conceivable future observations.

For composite hypotheses, pointwise equivalence at matched parameter values is insufficient for equal Bayesian evidence. The marginalized likelihoods must also agree. For example,

$$p(D \mid O_i, e) = \int p(D \mid \theta_i, O_i, e)\, \pi_i(d\theta_i \mid O_i, e)$$

can differ because the realization or parameter priors differ. Equal prior-predictive distributions give a Bayes factor of one wherever the common likelihood is nonzero; equal predictions at selected parameter settings need not. Prior preference remains possible, but must not be redescribed as an empirical likelihood advantage.

3.3 Representation equivalence

A structure- or probability-preserving map can establish equivalence for its specified domain. It does not automatically establish identical metaphysical interpretations, gauge redundancy, or equal explanatory value. Nor does a mathematically invertible reparameterization necessarily preserve which operations an observer can perform. An equivalence argument must state the common state space, physical inner product when relevant, observable map, and access restrictions. A failure of such an equivalence can be physically important without immediately deciding an ontology.

4 Case I: what a relativistic present would require

The elementary simultaneity result is exact within its standard domain. A boost along one spatial axis gives

$$t' = \gamma(t - vx/c^2), \qquad \gamma = (1 - v^2/c^2)^{-1/2}, \qquad |v| < c.$$

An event $(0, L)$ with $L \neq 0$ is simultaneous with the origin in one frame but has $t' = -\gamma vL/c^2 \neq 0$ for a nonzero boost. Thus that simultaneity hyperplane is not invariant under all boosts fixing the origin. This is standard kinematics, not a new result or a proof that nothing changes.

To obtain an ontological conclusion one must say why presentness must be that kind of hyperplane and respect the relevant symmetries. Putnam's argument invokes an observer-neutral reality-transfer principle; its broader closure argument includes further assumptions about observers (Putnam, 1967). Stein's event-relative approach supplies a different account of definite past structure (Stein, 1991). The present draft does not reproduce Stein's characterization theorem: the local audit found an isolated OCR statement too weak, while the original printed formulation remains unverified.

One response adds a preferred foliation. That is a substantive extra structure, not a free conclusion of SR. But observability also requires care. Appending a structure without changing old dynamics, probability assignments, or accessible operations need not change the old predictions. This conditional observation does not establish that any particular preferred-foliation theory is admissible or empirically equivalent; a coupling can make the structure observable. Nor does mere covariance of a notation establish absence of a physical preferred structure.

Another response disputes that simultaneity must provide the unity of what exists. Brading develops such a presentist program (Brading, 2013). Selected passages were inspected, but no quantitative discriminator has been established in this manuscript. That is enough to identify a premise needing argument, not enough to vindicate the whole program.

There are real philosophical costs on both sides. A preferred structure may be explanatorily idle; a local becoming relation may do no more than restate causal order; and a block account owes an account of the phenomena motivating passage. The point is to locate these costs accurately. A geometrical constraint on one global-present realization is not a completed comparison of all these packages.

The strongest schematic foliation reply is not that the structure must be detectable. It is that a primitive foliation constitutes which entities exist together, even if it changes no measured quantity. A critic can question its motivation or its explanatory return; merely noting observational redundancy does not answer a constitutive claim. The same standard applies to a local becoming bridge: it could be defended as specifying definiteness rather than causing a new detector response, but must explain what motivates that role instead of simply assigning a temporal name to the causal past. These are deliberately schematic rival arguments, not fully audited reconstructions of a particular defender. The comparative explanatory judgment remains open.

5 Case II: relational dynamics without an ontological shortcut

Relational clock constructions expose a second inference risk. A constrained global description can admit conditional evolution relative to a chosen clock. The relational-trinity work relates Dirac-observable, conditional Schrodinger, and reduced Heisenberg descriptions under specified hypotheses (Höhn et al., 2019). Its noninteracting setup and clock conditions matter. Interaction treatments already exist, including the kernel equation studied by Smith and Ahmadi (2019); "add interactions" is not a generic unoccupied research gap.

Two implications should be resisted. First, a stationary constraint equation does not by itself mean that subsystems cannot change: correlations can support conditional dynamics. Second, successful conditional dynamics does not by itself establish that correlations exhaust temporal reality. A reductionist argument needs a defensible sufficiency principle identifying all relevant temporal roles and showing that the construction realizes them. A primitive passage advocate cannot dismiss the construction, but may dispute that exhaustiveness premise.

The inverse mistake is equally important. An interaction-dependent or nonunitary conditioned evolution does not establish objective passage or an open future. One must specify the physical state space, clock instrument, normalization, accessible measurements, and what the proposed ontology would predict differently. A transformation that changes access restrictions cannot be treated as experimentally neutral merely because it is invertible on an enlarged mathematical space.

This leaves a substantive research task rather than universal agnosticism: find the precise boundary of a published equivalence under fixed operational conditions, then examine which philosophical bridge is strengthened or weakened. The boundary must first be checked against the interacting, periodic-clock, and quantum-reference-frame literature. No such new boundary is claimed in the present draft.

6 A finite record example and its exact limits

The following elementary construction makes one possible equivalence burden explicit. It is an expository application of ordinary probability, not a novel theorem about becoming.

Proposition 1 (Record-preserving translation)**.** *Let (X, μ) and (Y, ν) be probability spaces, and let $F : X \to Y$ be measurable with $F_*\mu = \nu$. The outcome kernels share one measurable outcome space. For each allowed complete observation protocol e, suppose $K_X^e(x, D) = K_Y^e(F(x), D)$ for μ-almost every x, for every outcome event D. Then the two packages assign the same probabilities to those protocols.*

Proof. For any declared e, D, the probability in X is

$$\int_X K_X^e(x, D)\, d\mu(x) = \int_X K_Y^e(F(x), D)\, d\mu(x) = \int_Y K_Y^e(y, D)\, d\nu(y),$$

which is the probability in Y. The result assumes rather than derives the measure and kernel preservation. □

For a finite horizon T and finite state space S, let

$$\mu(x_0, \ldots, x_T) = \pi(x_0) \prod_{t=0}^{T-1} P_t(x_{t+1} \mid x_t).$$

One implementation samples an initial state and then successive transitions; another samples an entire path with this law and releases its entries in order. Any common protocol restricted to the revealed prefixes, whose auxiliary randomness has the same conditional law given those prefixes and no access to the unrevealed path, yields the same record distribution. Adaptive choices of which revealed data to read can be included in the complete kernel. Operations modifying future transitions are not included unless their conditional dynamics are separately matched.

An exact numerical illustration. Take $S = \{0, 1\}$, $T = 2$, a fair initial state, and at each step a probability $3/4$ of staying in the same state. The path 000 has probability $(1/2)(3/4)(3/4) = 9/32$; 010 has probability $1/32$. Both implementations assign these same probabilities and the same prefix marginals. A companion standard-library check enumerates all paths using exact fractions. It verifies this example, not the general measure-theoretic argument or a theory of time.

Several limits are decisive. A sequential generator is not thereby the growing block theory; a sampled path is not thereby eternalism. The observer-access condition, the measure, and the identification of records do the operational work. If a different package gives access to unrevealed entries, or responds differently to interventions, the stated equivalence need not apply. Such extra access is a changed model assumption, not something entailed by one ontology label.

Observational agreement also falls short of interventional agreement. Consider a fair latent bit Z. In model A set $X = Z$ and $Y = X$; in model B set $X = Z$ and $Y = Z$. Both observational distributions have $X = Y$ with a fair marginal. Under an intervention replacing the assignment for X by $X := 0$, model A has $Y = 0$, whereas model B retains a fair Y. This familiar type of causal example shows why accessible interventions must be specified. It does not attribute either causal model to a temporal ontology.

The philosophical payoff is limited but useful: records cannot favor one implementation when measure and complete allowed response kernels are matched. To turn that into an argument about becoming requires a separately defended interpretation and a nontrivial comparison with prior work on histories and covariant growth. Calling the implementations two ontologies would skip exactly the premise the paper asks us to inspect.

7 Objections and replies

Bridge premises are unavoidable; your test is too demanding. Correct: every interpretation uses bridges. Their presence is not a defect. The demand is that the bridge be visible, defensible, and relevant to the contrast. A bridge can be constitutive of a view or supported by explanatory arguments. We reject the inference "extra premise, therefore arbitrary," as firmly as "physical equation, therefore metaphysical conclusion."

This makes every view compatible with physics. Only if admissibility is unconstrained, which the framework rejects. A view that discards its defining commitments, adds ad hoc structure, or fails empirical adequacy is not rescued by renaming the remainder. Conversely, showing one admissible realization establishes compatibility, not equal plausibility. A full exclusion claim has a stronger quantifier burden than a counterexample to one proposed proof of exclusion.

Relativity and parsimony already favor the block. They may favor a specified block package over a competitor with unjustified structure. The paper's demand is to articulate that explanatory comparison: which structure is avoided, whether the rival needs it, and what explanatory work competing packages perform. An inference to the best explanation is not discredited by being

philosophical rather than a direct detector reading. It is also not rendered deductively compulsory by the success of SR.

Phenomenology supplies evidence for passage. It supplies phenomena a satisfactory account should explain. It does not automatically choose among explanations. The advocate of primitive passage must show why experience supports that cause over alternatives; the critic must explain the experience rather than simply call it illusory. The accounts discussed by Callender (2017) and Skow (2015) are relevant comparators, not settled answers inherited by this draft.

An inaccessible ontology is physically meaningless. That conclusion needs a verificationist or other semantic argument which we have not assumed. Empirical equivalence can coexist with differences in grounding, explanation, simplicity, or modal commitment. The narrower conclusion is that matched observational likelihoods do not themselves select between the packages. Physics can still contribute through structural constraints and explanatory considerations.

Your record example is trivial and deliberately restrictive. The mathematics is elementary and the restrictions are deliberate. It is not offered as the novel contribution. Its purpose is to distinguish a proved conditional equivalence from an unsupported claim about all ontologies. A publishable extension would need a nontrivial physical class, a justified access model, and a result not already provided by the nearest predecessors.

8 What would turn this into a research contribution?

The present argument supplies a way of auditing claims, not a discovery that underdetermination exists. At least three developments could make it more than an organized restatement of familiar cautions:

1. A source-faithful reconstruction showing that a widely used inference requires a neglected bridge, and that the strongest published defenses and replies do not already establish the same point.

2. A nontrivial classification of when specified physical realizations preserve records and interventions, with explicit counterexamples at the boundary and a comparison to known equivalence results.

3. A substantive defense of one bridge principle that survives alternative temporal interpretations and explains why the resulting constraint is not merely a preference built into a definition.

The independent novelty gate is unmet. Neither the elementary proposition nor the use of agents, Python, or a future Lean proof supplies priority. Source audits, model reproductions, and the strongest contrary philosophical arguments are still needed. General relativity and cosmological time also require a separate extension: the existence of Cauchy slicings under global hyperbolicity (Bernal and Sánchez, 2005) is not uniqueness of a metaphysical present.

9 Provisional conclusion

Physics can make temporal metaphysics harder, more precise, and sometimes empirically vulnerable. It does not follow that every temporal distinction is physically idle; nor does a successful mathematical

representation settle all questions of existence and passage. The appropriate unit of comparison is a defended ontology–physics–bridge package with explicit access conditions. An argument should state whether it excludes such a package, discriminates its predictions, or establishes a representation equivalence. Keeping those conclusions distinct is a precondition for discovering, rather than assuming, what physics tells us about time.

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